ADAM D. RUSHDY

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EDUCATION

Purdue University - B.S. Aeronautical & Astronautical Engineering, Honors *Expected Graduation: May 2028*
GPA: 3.82/4.00
Activities: Purdue Aerial Robotics Team, Teaching Assistant, Purdue All-American Marching Band

TECHNICAL PROJECTS

R&D Airframe Member - *Purdue Aerial Robotics Team (PART)* *Sep 2025 - Present*
- Designed 9 wing structure and tail assembly components for a team eVTOL aircraft in SolidWorks, incorporating tolerance and fit considerations for a modular, ease-of-assembly build.
- Performed structural/stress analysis in MATLAB to evaluate rib spacing trade-offs between panel buckling and structural weight margins, informing airframe design decisions.
- Developed CAM toolpaths for CNC machining of 2 precision carbon-fiber layup molds for wings within a 2-hour manufacturing constraint using Fusion 360.
- Fabricated 2 composite wing assemblies using carbon fiber layup over laser-cut internal structures.

UAV Inertial Navigation System *June - July 2026*
- Developed an INS with Python using ESP32, IMU, GPS, and magnetometer fusing 9 sensor streams via quaternion-based 6-DOF complementary and Kalman filtering at 40 Hz for real-time state estimation.
- Reduced position RMSE from 13 m to 4.9 m (62% improvement) by diagnosing attenuation of high-dynamics events and re-tuning Kalman noise covariances, benchmarked against ArduPilot EKF logs to achieve <5 m error.

Fixed-Wing Surveillance UAV - Design, Analysis & Fabrication *Mar - June 2026*
- Designed and fabricated a 1.3 m, 1.8 kg fixed-wing UAV with Raspberry Pi camera integration. Selected airfoils in XFOIL, sized aircraft weight and geometry for 25 m/s cruise, and modeled the full assembly in SolidWorks.
- Designed wing structure for a 6-g limit load factor, used FEA to confirm hand-calculations & structural integrity with a maximum tip deflection of 12.1 mm. Fabricated using carbon-fiber skin layups over 3D-printed internal ribs and spars.
- Performed aerodynamic, stability, and flight dynamics analysis in XFLR5, iterating on CG, tail moment arm, and wing geometry to achieve a 6.67% static margin.
- Implemented ArduPilot flight controller for fly-by-wire and data logging. Completed maiden flight capturing airspeed, load factor, battery performance, and estimated position across a 150-second flight, validating the ~25 m/s cruise speed and maximum 3-g load against design predictions.

WORK EXPERIENCE

Engineering Intern - *Xchanger* *Hopkins, MN* | *May - Aug 2026*
- Diagnosed and troubleshot a production model error causing predicted pressure-drop to underestimate actual values by 2-3x, preventing inaccurate data sheets from reaching clients.
- Tested pressure-drop and velocity performance across 18 air-to-air heat exchanger units (100% of models) and multiple flow configurations, generating an empirical dataset used to recalibrate the company's C-based heat exchanger model.
- Derived unit-specific correction factors now built into the production model, giving engineers more accurate datasheets and a validated safety margin when quoting new units.

Teaching Assistant - *Honors Intro to Engineering (ENGR 16X)* *West Lafayette, IN* | *Aug 2025 - Present*
- Support 60+ first-year students with coursework and grading, including 6 design projects spanning Python modeling, MATLAB simulation, and Raspberry Pi embedded systems.

Intern - *Seminar for Top Engineering Prospects (STEP)* *West Lafayette, IN* | *Jun - Aug 2025*
- Developed and delivered a modular Arduino programming curriculum to 320 prospective engineers, built functions and tutorials to teach students basic programming skills.

Teaching Assistant - *Intro to C Programming* *West Lafayette, IN* | *Jan - May 2025*
- Led 30+ students across 12 labs covering core C concepts, held office hours, and graded coding assignments.

SKILLS

CAD / Analysis: SolidWorks (CAD/FEA), Fusion 360 (CAD/CAM), NX, Teamcenter, XFLR5, XFOIL
Programming: MATLAB/Simulink, Python, C/C++, Java, Arduino, Git/GitHub
Fabrication: Carbon-fiber composite layup, 3D printing (FDM), CNC milling
Leadership: Eagle Scout, Assistant Snare Segment Leader